Conditional probability NJS

1. Two fair dice are rolled. Given that they show different numbers, what is the probability that one of them shows a six?

2. Each question on a multiple choice test has five possible answers. When a student takes this test, model the probability that they know the answer to a question as $\frac{1}{3}$. If the student does not know the answer they guess at random from the available choices. Given that the student answers a question correctly, what is the probability that they knew the answer?

3. Urn A contains 2 white and 4 red balls. Urn B initially contains 1 white and 1 red ball. A ball is chosen at random from urn A and put into urn B. You then chose a ball at random from urn B. What is the probability that:
   a. the ball selected from urn B is white;
   b. the transferred ball is white, given that the ball chosen from urn B is white?

4. Die A is fair, and rolls a 6 with probability $\frac{1}{6}$. Die B is weighted so that a 6 is rolled with probability $\frac{1}{3}$. A die is chosen at random and rolled twice. If both rolls are a 6, then what is the probability that the weighted die was chosen?

5. A gambler has in his pocket a fair coin and a two-headed coin. He selects the one coin at random; when he flips it, it shows heads.
   a. What is the probability that it is the fair coin?
   b. Suppose that he flips the same coin a second time and again it shows heads. What now is the probability that it is the fair coin?
   c. Suppose that he flips the same coin a third time and it shows tails. What is now the probability that it is the fair coin?
   d. Suppose that he flips the same coin a fourth time and it shows heads. What is the probability that it is the fair coin?

6. An insurance company believes that people can be divided into two classes: those who are accident prone and those who are not. They believe that 30 percent of the population is accident prone. Their statistics show that an accident-prone person will have an accident at some time within a fixed 1-year period with probability 0.4, whereas this probability decreases to 0.2 for a non-accident-prone person.
   a. What is the probability that a new policyholder will have an accident within a year of purchasing a policy?
   b. Suppose that a new policyholder has an accident within a year of purchasing a policy. What is the probability that he or she is accident prone?
   c. What is the probability that someone has an accident in the second year, given that he or she had an accident in the first year?

7. At a certain stage of a criminal investigation the inspector in charge is 60% convinced that the prime suspect is guilty. Suppose now that a new piece of evidence is discovered that shows that the perpetrator has a certain characteristic (like having black hair). If the suspect, along with 20% of the population, has this characteristic, then how convinced of the suspect's guild should the inspector now be?

8. A deck of cards is well-shuffled, and the top three cards are revealed one at a time. If the second and third cards are spades, then what is the probability that the first card was a spade?

9. You have three cards which are identical except that one card is red on one side and green on the other, one card is red on both sides, and one card is green on both sides. The three cards are put into a bag, and one is drawn at random and placed on the ground. Given that the face showing up is red, what is the probability that the chosen card is red on both sides?

10. When repeatedly rolling a die, what is the probability that you roll a 6 before you roll a 1?

Bonus problems

11. An ordinary deck of 52 cards is well-shuffled and the cards are then turned over one at a time until the first ace appears. Given that the first ace is the 20^{th} card to appear, what is the probability that the card following it is the:
    a. ace of spades;
    b. two of clubs?

12. A deck of cards is shuffled and then divided into two halves of 26 cards. A card is drawn from one of the halves; it turns out to be an ace. The ace is then placed in the second half-deck. That half is then shuffled, and a card is drawn from it. What is the probability that the card drawn is an ace?

13. Consider the following game. A deck of cards is shuffled and its cards are turned face up one at a time. At any time you can shout "next" and if the next card is the ace of spades, then you win, and if not, then you lose. If the ace of spades appears before you shout "next", then you lose. If there is only one card remaining, and you have never said "next", then you win automatically.

    Show that whatever strategy you employ for deciding when to shout "next", you probability of winning is $\frac{1}{52}$ (and so you might as well just say next before the first card is turned over, and get the game over and done with…)

14. A parallel system functions whenever at least one of its components works. Consider a parallel system of n components and suppose that each component works with probability $\frac{1}{2}$. Given that the system is functioning, find the probability that component 1 is working.

15. The colour of a person's eyes is determined by a single pair of genes. If they are both blue-eyed genes, then the person will have blue eyes; if they are both brown-eyed genes, then the person will have brown eyed; and if one of them is a blue-eyed gene and the other is a brown-eyed gene, then the person will have brown eyes. A newborn child independently receives one eye gene from each of its parents and the gene it receives from a parent is equally likely to be either of the two eye genes of that parent.

    Suppose that Tina, and both of her parents have brown eyes, but that Tina's brother Eugene has blue eyes
    a. What is the probability that Tina possesses a blue-eyed gene?

    Suppose that Tina procreates with a blue-eyed man.
    b. What is the probability that their first child will have blue eyes?
    c. Given that their first child has brown eyes, what is the probability that their second child will also have brown eyes?

16. A pair of dice are repeatedly rolled. Find the probability that an even total is rolled 6 times before a total of seven is rolled.

17. Four people sit around a table. Person 1 throws a die. If they roll a 6 they are the winner; otherwise they pass the die to Person 2, who throws the die. If person 2 rolls a 6 they are the winner; otherwise they pass the die to Person 3, who throws the die. If person 3 rolls a 6 they are the winner; otherwise they pass the die to Person 4, who throws the die. If person 4 rolls a 6 they are the winner; otherwise they pass the die to Person 1, and the whole process is repeated.
    Find the probability that person 3 wins the game.